

Economic Downturns and Political Competition since the 1870s

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Relying on new data on the ideology of heads of government in 27 democracies over a period of more than 140 years, this article shows that short economic downturns, with a single year of falling per capita consumption, have more often resulted in shifts to the right than shifts to the left. But long-lasting economic downturns, with more than one consecutive year of falling consumption, are different, since they tend to affect a much greater proportion of the population: compared with short downturns, which favor the right, long downturns have more uniform political effects.

It is well known that economic downturns harm governing parties. Less is known about their ideological effects: their consequences for the struggle between parties on the left and parties on the right. Several studies of public opinion and electoral behavior in the advanced democracies in the post-war period suggest that downturns should favor the right, but the liberal and social-democratic breakthrough in the 1930s, in the wake of the Great Depression, suggests otherwise, and the relationship between economic downturns and political competition before the First World War has not been studied systematically.

This article argues that brief episodes of economic contraction (“short” downturns) and drawn-out periods of economic decline (“long” downturns) have different ideological effects: short downturns tend to favor parties on the right, but the political consequences of long downturns are more uniform. A short downturn typically only affects marginal groups. The main concern of pivotal, middle-income voters, therefore, is that increasing demands for social protection and redistribution might lead to higher taxes, which means that the net result of a short downturn is a shift to the right. A long downturn, by contrast, typically affects a greater proportion of the population, which counteracts the first effect and increases the likelihood of a shift to the left.

This argument was first made in Lindvall (2014) on the basis of evidence from the Great Depression and the Great Recession. In this article, I develop the argument further and show that it reaches far beyond the two individual crises that

I examined in my previous article. Using new data on the ideology of heads of government in 27 democratic states during the past 140 years, this article analyzes switches between center-left and right-wing heads of government and finds that short economic downturns increase the likelihood of shifts to the right but not the likelihood of shifts to the left, whereas long economic downturns increase the likelihood of a change in government ideology regardless of the incumbent government’s ideology. Additional empirical analyses that also take the depth of economic downturns into account find that the likelihood of a shift to the left only increases during downturns that are both long and deep; all other types of downturns have historically been more likely to result in a shift to the right.

As far as I know, this article provides the first cross-country comparative analysis of the relationship between economic downturns and the political competition between the left and the right that covers a period of more than 100 years. The findings, which suggest that there is a systematic relationship between economic downturns and ideological competition in democracies, have important implications for theories of comparative political economy and economic voting.

ECONOMIC DOWNTURNS AND POLITICAL COMPETITION

The idea that an economic downturn is more likely to lead to a shift to the right than a shift to the left, which we can call the *swing-to-the-right hypothesis*, goes back to studies

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Data and supporting materials necessary to reproduce the numerical results in the paper are available in the JOP Dataverse (<https://dataverse.harvard.edu/dataverse/jop>). An online appendix with supplementary material is available at <http://dx.doi.org/10.1086/692787>.

by Alt (1979), who argued that macroeconomic problems rendered voters in the United Kingdom less “altruistic”; Durr (1993), who argued that voters in the United States become more liberal when the economy is growing and more conservative when it is not; and Stevenson (2001), who concluded, on the basis of cross-country comparative evidence, that a weak economy tends to push the electorate’s policy preferences rightward. Alt’s, Durr’s, and Stevenson’s arguments are slightly different from each other, but the underlying idea is the same: there is something about economic downturns that renders large groups of voters less supportive of high levels of social spending, taxation, and redistribution. In other words, economic downturns make voters less likely to support the sorts of policies that are typically favored by left-wing and progressive-liberal parties.¹

The opposite view can be called the *swing-to-the-left hypothesis*. According to this view, economic downturns increase the demand for social spending and redistribution, and, consequently, the support for left-wing and progressive-liberal parties (see the discussion in Stevenson [2002]). Unlike the studies that have found evidence for the swing-to-the-right hypothesis—which have typically been based on analyses of survey data stemming from the second half of the twentieth century—most of the evidence for the swing-to-the-left hypothesis comes from macro-level analyses of historical political realignments, especially those of the 1930s. Peter Gourevitch (1984; 1986, chap. 4) famously argued, for instance, that the Great Depression resulted in a political realignment based on “red-green coalitions” between workers and farmers in countries such as Sweden and the United States. The swing-to-the-left hypothesis receives additional support from the literature on issue ownership, which tends to find that left-wing parties benefit politically from high levels of unemployment (see, e.g., Wright 2012; cf. Dassonneville and Lewis-Beck 2013; Kiewiet 1983).

These arguments about the ideological effects of economic downturns can be contrasted with the *punishment hypothesis*, which follows straightforwardly from the dominant model in the literature on the political consequences of economic events: the retrospective-economic-voting model. The main implication of the economic-voting model is that governing parties do badly in economic downturns, regardless of the ideological orientation of those parties. Countless studies have indeed shown, using data that go back to the interwar period, that incumbents are more likely to lose votes in hard times than in good times (Anderson et al. [2004], Duch and Stevenson 2008, Lewis-Beck [1988], and van der Brug, van der

Eijk, and Franklin [2007] are some of the major comparative studies of economic voting from the past 30 years).²

Why long downturns are different

This article argues that brief episodes of economic contraction—“short” economic downturns—are more likely to result in a shift to the right than a shift to the left, whereas drawn-out, “long” economic downturns have more uniform political effects (they increase the likelihood of ideological shifts regardless of the incumbent government’s ideological orientation).

The argument goes as follows. In a short economic downturn—or in the beginning of a long downturn—pivotal, middle-income voters are typically not personally affected, which means that the kinds of mechanisms that scholars such as Alt, Durr, and Stevenson have described are likely to dominate: knowing that an economic downturn increases the demand for social protection and redistribution from those who are personally affected, pivotal voters fear higher taxes and become less “altruistic,” as Alt claims, or regard left-of-center policies as “luxury goods,” as Stevenson argues. The net result is a shift to the right in the electorate. There is evidence for this claim about the short-term effects of economic downturn from the Great Recession, the economic crisis that began in 2007–8: Margalit (2013, 99), who has examined social policy preferences in the United States during the first years of the Great Recession, concludes that the crisis led to a “bifurcation in sentiment between the narrower constituency who personally experienced a major economic setback and the broader population that did not,” which meant that the net effect was a short-term shift to the right.³

If an economic downturn proves to be more long-lasting, however, it typically affects a greater proportion of the population, either directly or indirectly. This counteracts the short-term shift to the right, leveling the political playing field. The economic harm caused by a long downturn is not contained within marginal groups. This means that the proportion of voters that demand social protection, redistribution, and higher government expenditures increases, whereas the proportion of voters that resist such demands since they are not personally affected decreases. During a long-lasting economic downturn,

1. The beginning of this section follows the discussion in Lindvall (2014).

2. On economic voting in the Great Depression, see Achen and Bartels (2005) and King et al. (2008); on economic voting in the Great Recession, see, for instance, Bartels (2014), Hernández and Kriesi (2016), Kriesi (2014), and the contributions to the two special issues of *Electoral Studies* that deal with the consequences of the crisis (vol. 31, no. 3 [2012] and vol. 32, no. 3 [2013]).

3. Lindgren and Vernby (2016) also find evidence of a shift to the right in the first two years of the Great Recession in Swedish district-level data.

in which economic vulnerability begins to be felt across many different social groups, the center-left can thus be expected to do better than it does during shorter economic downturns—particularly if the downturn is deep enough to cause widespread economic harm when the effects of the downturn accumulate over time. There is evidence for this claim about the difference between the short-run and long-run political effects of economic downturns from the Depression era. Kenworthy and Owens (2011, 218) argue, for instance, that the ideological shift to the left in the 1930s in the United States, and elsewhere, occurred precisely because the Depression was so long-lasting: “In the early years of the Great Depression it was not clear that large changes in Americans’ attitudes toward politics, fairness and government activism were under way,” Kenworthy and Owens observe, “But in the end the Depression did contribute to enduring shifts.”

The argument that we need to distinguish between different kinds of downturns when we examine the political effects of macroeconomic events can be seen as an attempt to reconcile the swing-to-the-right hypothesis and the swing-to-the-left hypothesis. In my view, scholars such as Alt, Durr, and Stevenson correctly observe that the short-run effect of an economic downturn is to shift the political preferences of many voters to the right. But the lesson from drawn-out economic crises in the past is that over time, this short-run effect is counteracted by a medium-to-long-run effect that is quite different: when the part of the electorate that is personally affected by a downturn becomes larger and more heterogeneous, demands for social protection become stronger, center-left electoral coalitions such as the “red-green” coalitions of the 1930s become more likely, and the prospects for left-wing and progressive-liberal parties improve.

One possible objection to this argument is that even a brief episode of economic contraction may affect a large proportion of the population if the drop in economic output and consumption is big enough. I agree that the depth of an economic downturn matters, in addition to its duration. But it almost always takes time before broad groups of middle-income voters are harmed by a downturn—even a deep one—since employment contracts typically cannot be terminated instantaneously, since the knock-on effects of a downturn in one sector only reach other sectors with some delay (see below), since wages and prices are “sticky,” since labor market “outsiders” suffer sooner than “insiders” (Rueda 2007), and since core workers are often protected by unemployment insurance (Western Europe and North America) or severance-pay programs (Latin America) in the first phase of an economic crisis. I therefore expect that only long-lasting downturns will have economic effects that are sufficiently widespread to result in a level political playing field or an increased likelihood of a

shift to the left. In all other cases, I expect that recessions will do greater harm to the prospects of center-left parties than to the prospects of parties on the right.

This explanation of the relationship between economic downturns and political outcomes combines retrospective-voting mechanisms and prospective-voting mechanisms. Retrospective economic voting matters: my argument assumes, and my results show, that on average, incumbents are harmed by an economic downturn. But prospective voting also matters, since the policies that voters desire in an economic downturn are different from the policies that they desire in normal times: voters that are harmed by an economic downturn want more redistributive policies and more social protection, but voters that are not harmed are worried that taxes will rise and therefore shift their policy preferences rightward. The decisions that voters ultimately make are shaped by both retrospective and prospective considerations.

The claims that I have put forward here only follow from the underlying theoretical argument if long downturns cause more widespread harm than short downturns. I would therefore like to spend some time justifying this key assumption.

I start with the effects of economic downturns on unemployment, since rising unemployment is typically the most apparent—and important—effect of an economic downturn, from the point of view of the voters. No cross-country-comparable data on unemployment are available for the period before 1960. In the first two columns of table 1, I therefore present estimates of the mean change in the unemployment rate—relative to the year before the downturn—for every economic downturn that occurred in the period between 1960 and 2009 in the 23 OECD countries that are included in the Armingeon et al. (2011) comparative-politics data set (relying on the main consumption-based indicator of eco-

Table 1. Change in Unemployment over Year before Downturn

	OECD 1960–2009		United States 1890–2009	
	ΔU	N	ΔU	N
1 year of negative growth	+ .6	92	+ .6	15
2 years of negative growth	+ 1.8	35	+ 3.4	5
3 years of negative growth	+ 4.1	11	+ 7.2	3
4 years of negative growth	+ 9.0	3	+ 9.3	2

Note. Data source, economic downturns: Barro and Ursúa (2008). Data source, unemployment, OECD: Armingeon et al. (2011). Data sources, unemployment, United States: Coen (1973), Romer (1986), Bureau of Labor Statistics. US estimates exclude the two world wars.

nomic downturns that I define below). As the table shows, during one-year economic downturns—or during the first year of longer downturns—the unemployment rate increased, on average, by less than one percentage point in the advanced industrialized democracies in this 50-year period. In the second year of longer downturns, however, the average increase in unemployment was 1.8 percentage points, relative to the year before the downturn; in the third year, the average was 4.1 percentage points, and in the fourth year (a rare occurrence), the average increase in unemployment was 9 percentage points.

For the United States, it is possible to construct a reasonably consistent series of unemployment-rate estimates for a much longer time period: Romer (1986) has compiled a series for the period from 1890 to 1930, Coen (1973) presents unemployment-rate estimates for the period from 1922 to 1940, and the Bureau of Labor Statistics has data for the period from 1944 to the present. Evidence from these US data can be found in the last two columns of table 1. The US pattern for 1890–2009 is very similar to the OECD pattern for 1960–2009: during long-lasting downturns, the increase in unemployment has typically been very large; the unemployment increase during one-year downturns (or in the first year of longer downturns) has been comparatively small. The evidence is clear: the effects of a long downturn are felt by a greater proportion of the population than the effects of a short downturn.

But these labor-market effects are not all. As I mentioned earlier, one of the arguments that scholars such as Gourevitch (1984) have made about the Great Depression is that the economic crisis of the 1930s resulted in new electoral coalitions, typically on the center-left, between groups of voters whose economic interests were not previously aligned (such as workers and farmers, who formed new “red-green” coalitions). The reason was that the Great Depression did not only harm workers; it had many other harmful economic effects as well. Case-study evidence from long economic downturns suggest that this is a common pattern: long downturns typically have several different types of economic effects, affecting different groups of voters.

Most importantly, long historical downturns have often been associated with crises in the financial sector and with falling asset prices (notably depreciating house prices). This connection between asset prices and real economic decline was an important element of the Great Depression in the United States, during which, as Bernanke (1983, 15, 17) notes, many borrowers, especially “households, farmers, unincorporated businesses, and small corporations,” suffered from the “progressive erosion of . . . collateral relative to debt burdens.” In other words, the economic harm that the Depression did was not limited to those who lost their jobs. The same connection between financial-sector crises, falling asset prices,

and falling output can be observed in more recent economic crises such as the uncommonly deep and long-lasting crises in Finland and Sweden in the early 1990s, which resulted in similarly widespread economic harm (Jonung, Kiander, and Vartia 2009).

Hypotheses and contributions

On the basis of the arguments and evidence that I have discussed in this section, I expect to find (a) that short economic downturns are associated with a significantly higher likelihood of a shift to the right if the incumbent government is left-wing or centrist; (b) that there is, at most, a weak relationship between short economic downturns and the likelihood of a shift to the left if the incumbent government is right-wing; and (c) that long downturns have more uniform political effects, in the sense that the likelihood of a change in government ideology increases irrespective of the incumbent government’s ideology.

Previous studies of the relationship between economic downturns, election outcomes, and changes in political leadership have typically generated inconclusive results. Stevenson (2002)—one of the few published studies that have sought to describe the effect of economic downturns on election outcomes—finds some support for the swing-to-the-right hypothesis but also observes that the results are unstable (for a more recent study of election results in the postwar period, see Grafström and Kayser [2014]). One of the implications of the ideas and evidence that I present in this article is that by taking the distinction between “short” and “long” economic downturns into account, we should be able to generate more precise and stable estimates of the empirical relationship between economic downturns and macro-level political outcomes.

The empirical analysis in this article makes few assumptions about the level of knowledge that voters have about politics and the economy. Since the outcome variable is a simple indicator of the ideology of the head of government (the president, prime minister, or chancellor), this study only assumes that voters know the ideological orientation of one of the country’s most visible elected political figures, and since the main explanatory variable is a simple indicator of negative per capita consumption growth, this study only assumes that voters understand whether the economy is shrinking (and, if so, whether it has been shrinking for one year or more).

This article also covers a comparatively long time period (most other studies are based on postwar data; I include the interwar period and, for some countries, the period before the First World War), and its geographical scope is broad (most previous studies deal exclusively with the rich de-

mocracies in North America, Western Europe, and the Asia-Pacific region; I include several countries in Latin America).

This broad temporal and geographical scope comes at a price, however. Since data on potential confounders are scarce when we go back to the period before the Second World War, the empirical models are relatively weakly specified, and in the absence of individual-level data on preferences or even election outcomes, it is not possible to examine the micro-level mechanisms behind the observed relationship between economic downturns and changes in government ideology. Moreover, the connection between election outcomes and the appointment of heads of government is not straightforward in multiparty systems (for instance, it is quite possible for the head of government's party to increase its vote share but still lose power), so changes in the ideology of the head of government are not always correlated with changes in the distribution of votes and seats among political parties.

As I have already mentioned, the argument of this article is closely related to the argument in Lindvall (2014). In my earlier article, I compared the Great Depression of the 1930s with the Great Recession of the 2000s, showing that the electoral consequences of these two events were surprisingly similar: in both periods, right-wing parties were at first more successful than left-wing parties, but this effect only lasted for a few years, after which left-wing parties did better than right-wing parties. So far, I have developed my earlier argument by discussing what makes "long" economic downturns different from "short" economic downturns and by discussing the relationship between the duration and depth of economic downturns. The main contribution of this new article, however, is empirical. The evidence that I now present covers a much larger number of observations than Lindvall (2014) (all democratic elections since 1870 in 27 countries, as opposed to two elections per country). This article therefore suggests that the pattern that I found in my 2014 article is a general feature of political competition in democratic systems; it is not specific to the Great Depression and the Great Recession.

RESEARCH DESIGN AND DATA

The empirical analyses in this article are based on a new data set that provides information about the ideological orientation of heads of government in 32 countries from 1870 onward. The main methodological challenge for a study that seeks to capture ideological similarities and differences among political leaders in that many countries over a period of almost a century and a half is to develop a measure of ideology that can realistically be applied across regions and over time. I deal with this problem by separating heads of government into two distinct ideological categories: on the one hand the political left and center; on the other hand the political right. These

ideological labels correspond to party families that have existed in all of the countries in the sample throughout the period that I study: "left or center" denotes socialist, social-democratic, centrist-agrarian, and social-liberal parties and factions, as well as some centrist Christian democrats; "right" denotes conservative, Catholic, right-wing Christian democratic, and market-liberal parties and factions.⁴

It would in principle have been possible to distinguish between "left" and "center," creating three separate ideological categories instead of two, but relying on a binary outcome variable has several advantages. Most importantly, whereas right-wing parties or factions have existed in all countries in the sample throughout the period covered in the article, and whereas the proportion of right-wing heads of government in this set of countries in any given year has rarely been lower than 35%–40% (the overall mean is approximately 45%), there are many examples of countries that lack either competitive left-wing parties or competitive centrist parties—not least because left-wing governments were rare in the period before the First World War and centrist governments have been relatively rare since the late interwar period. Concentrating on the competition between parties on the left and the center on the one hand and parties on the right on the other therefore makes the data comparable across space and time.

The available data on per capita consumption and economic output are annual (quarterly and monthly data are only available for the last few decades). The empirical parts of the article therefore seek to answer the following question: does negative economic growth during an election year increase the likelihood that the ideology of the head of government changes during that year? In most cases, I compare the ideology of the head of government on January 1 of the year after the election with the ideology of the head of government on January 1 of the election year. Where elections are held in the late autumn, however, incoming heads of government often begin their terms in office after January; in these cases, I concentrate the incoming head of government (for example, although Herbert Hoover was still president on January 1, 1933, I count Franklin D. Roosevelt as president in 1933).⁵

Since the arguments that I developed in the theory section are only relevant for democratic political systems, I exclude all nondemocracies from the sample (relying on regime data from Boix, Miller, and Rosato [2013] to distinguish democ-

4. The appendix provides details regarding the coding of the data, which are based on a wide range of comparative and country-specific sources and have been cross-checked with country experts.

5. US presidents have been inaugurated in late January since the middle of the 1930s; before then, they were inaugurated in early March.

racies from nondemocracies). In other words, I do not model transitions to authoritarianism or transitions to democracy. Moreover, since the argument of the article is only relevant for changes in political leadership that occurred as a result of elections, I only include election years. To identify election years, I rely on the Przeworski et al. (2013, 17) data set, which provides data on elections in which “the office of the chief executive was at stake.”⁶

With the exception of Switzerland—which is excluded from the sample since it is a country with a semi-permanent, collective executive—the head-of-government data set that I examine in the article covers almost all North American, South American, and Western European countries with a current population of more than one million, plus five additional countries: Argentina, Australia, Austria, Belgium, Bolivia, Brazil, Canada, Chile, Colombia, Costa Rica, Denmark, Ecuador, Finland, France, Germany (West Germany between 1949 and 1990), Greece, Ireland, Italy, Japan, Mexico, the Netherlands, New Zealand, Norway, Paraguay, Peru, Portugal, Spain, Sweden, the United Kingdom, the United States, Uruguay, and Venezuela. In the main analyses in this article, I exclude Bolivia, Costa Rica, Ecuador, Ireland, and Paraguay because of missing data on consumption per capita (but these five countries are included in the GDP-based analyses that are reported in table 7 in the appendix, available online). The heads-of-government data set covers the period from 1870 (or later, for countries that were not sovereign in 1870) to 2012. Most of the empirical analyses in the article cover the period between 1872 and 2010 (excluding the two world wars). I start in the early 1870s since the rate of industrialization increased sharply in most of the countries in the sample around that time (Finer 1997, vol. 3, chap. 12). Political conflicts in pre-industrial societies were clearly different from those in industrial societies (as argued forcefully by Lipset and Rokkan [1967]), so I do not expect that economic downturns had the same types of effects in the pre-industrial world.

My main measure of economic downturns is based on annual data on consumption per capita that have been compiled by Barro and Ursúa (2008). Barro and Ursúa have updated Angus Maddison’s widely used historical data on economic output and added data on consumption. Since voters are personally affected by declining consumption but only indirectly affected by declining output, a measure of consumption per

capita is preferable to a measure of output per capita for the purposes of this study. Moreover, many of Barro and Ursúa’s updates and corrections concern the onset and duration of macroeconomic crises, which is an additional reason to rely on their data set.

I define a single year of declining per capita consumption as a “short” downturn. All downturns that last two years or longer consequently count as “long.” It would in theory be possible to develop more fine-grained measures of the duration of downturns, but few downturns last more than two years, so I combine all downturns lasting two years or longer into a single category.⁷

DESCRIPTIVE EVIDENCE

The sample that I examine excludes the First World War, the Second World War, years when no elections involving the office of head of government were held, and years when the head of government’s ideology is coded as “other.” This leaves us with a sample of 556 observations (some of the analyses that are based on GDP per capita data instead of consumption per capita data include a slightly larger number of observations). Seventy-five of the observations are from the period before the First World War; 88 are from the interwar period; 393 are from the postwar period. Sixty-three of the observations are from Latin America; 493 are from the other regions in the sample (Western Europe, North America, and the Asia-Pacific). Fifty-five percent of all incumbent governments are coded as left-wing or centrist; 45% are coded as right-wing. The ideology of the government changed from left or center to right or vice versa in slightly less than one-third of all cases (169). There are slightly more shifts to the left (89) than shifts to the right (80) since right-wing governments were more common in the beginning of the period than at the end.⁸

Consumption per capita fell in the year of the election in 24% of all cases. Twenty-five percent of the observed election-year downturns were “long,” in the sense that they started already before the election year. To be exact, there are 101 short downturns and 34 long downturns in the sample. Economic downturns were less frequent in the postwar period than in the period before the Second World War.

Table 2 describes the relationship between economic downturns and the rate of ideological change—that is, the proportion of all election years in which the ideology of the

6. I have made a few corrections in the coding of election years since I found that the information in the original data set was incorrect for a small number of observations. I also exclude a small number of observations in which either the incumbent or the incoming government’s ideological orientation was coded as “other” (almost all of these heads of government led nonpartisan, “technocratic” governments).

7. In table 7 in the appendix, I demonstrate that different ways of describing and distinguishing between “short” and “long” downturns lead to substantively similar results.

8. All the 169 observed shifts in the ideology of the government in connection with elections are listed in the appendix.

Table 2. Economic Downturns and Ideological Change: Descriptive Evidence

	Rate of Ideological Change (Percent) If ...			Increase in Rate of Change If ...		N
	No Downturn	Short Downturn	Long Downturn	Short Downturn	Long Downturn	
All countries and periods:						
Left/centrist incumbent	20.3	44.6	40.0	24.3	19.7	307
Right incumbent	33.7	35.6	64.3	1.9	30.6	249
Before First World War:						
Left/centrist incumbent	30.4	30.0	33.3	−.4	2.9	36
Right incumbent	36.4	28.6	66.7	−5.8	30.3	39
Interwar:						
Left/centrist incumbent	26.1	41.7	66.7	15.6	40.6	38
Right incumbent	39.3	26.7	71.4	−12.6	32.1	50
Postwar:						
Left/centrist incumbent	18.4	50.0	35.7	31.6	17.3	233
Right incumbent	32.1	50.0	50.0	17.9	17.9	160
Europe, North America, Asia-Pacific:						
Left/centrist incumbent	19.4	44.4	38.9	25.0	19.5	274
Right incumbent	29.8	33.3	66.7	3.5	36.9	219
Latin America:						
Left/centrist incumbent	30.0	45.5	50.0	15.5	20.0	33
Right incumbent	63.6	50.0	50.0	−13.6	−13.6	30

government changed. In the first two rows of table 2, we find that in the full sample, with all countries and periods included, short economic downturns were associated with a large increase in the proportion of left-wing and centrist governments that were replaced by right-wing governments (+24 percentage points), but only with a small increase in the proportion of right-wing governments that were replaced by left-wing and centrist governments (+2 percentage points). Long downturns, by contrast, were associated with a large increase in the rate of ideological change regardless of the incumbent government's ideology (larger, in fact, if a right-wing government was in power).

These patterns are fairly stable across time and space, at least when it comes to the apparent effects of short economic downturns (long downturns are rare events, so we are left with too few observations to draw any firm conclusions from the period-specific and region-specific data). As table 2 shows, short economic downturns are associated with a larger increase in the rate of ideological change if the government is left-wing or centrist than if the government is right-wing in all the subsamples (in some subsamples, economic downturns are in fact associated with a decrease in the rate of ideological change if the government is right-wing). When growth is positive, by contrast, the rate of ideological change is lower for

left-wing and centrist governments than for right-wing governments in all the subsamples.

The results in table 2 provide preliminary evidence for the main claims of this article: short economic downturns appear to be associated with a higher likelihood of a change in the ideology of the government when the incumbent government is left-wing or centrist, but not when the incumbent government is right-wing. Long downturns, on the other hand, appear to have more uniform political effects: during long downturns, the likelihood of ideological change seems to increase sharply irrespective of the incumbent government's ideology.

STATISTICAL EVIDENCE

In this section, I use statistical methods to estimate the likelihood of ideological change between the beginning of year $t - 1$ (the year of the election) and the beginning of year t (the year after the election) conditional on economic circumstances in year $t - 1$ (and earlier), and conditional on the ideology of the government in the beginning of the year $t - 1$. Since the main objective of this study is to determine if economic downturns have different effects on the likelihood of a change in the ideology of the government de-

pending on the ideology of the incumbent government (I expect short downturns to have stronger effects on shifts to the right than on shifts to the left), I estimate these effects separately, using transition models that have the following general forms (cf. Beck 2008; Diggle et al. 2002, 195):

$$\Pr(y_{i,t} = 1 | y_{i,t-1} = 0) = \text{Logit}(\beta x_{i,t-1}) \quad (1)$$

and

$$\Pr(y_{i,t} = 0 | y_{i,t-1} = 1) = \text{Logit}(\alpha x_{i,t-1}), \quad (2)$$

where $y_{i,t}$ is the ideology of the head of government of country i in year t (that is, on January 1 of the year after the election; $y = 1$ if the government is right-wing); $y_{i,t}$ is the ideology of the head of government of country i in year $t - 1$ (that is, on January 1 of the election year); and x are the explanatory variables (which are measured during year $t - 1$ the year that passed between the two observations of the outcome variable).

We can think of the outcome variables in models 1 and 2 as “ideological change”: the appointment of a head of government with a different ideology than the incumbent. An alternative approach would have been to define the outcome variable “(head of) government is right wing,” but the choice of outcome variable is largely a matter of taste, for this alternative approach is mathematically equivalent (the predicted probabilities that I report in table 4 and table 6 can also be derived from an alternative model that treats “(head of) government is right wing” as the main outcome variable).

To test whether recessions have different effects on the likelihood of a change in government ideology depending on the ideology of the incumbent government, these effects need to be estimated jointly. I do that by combining equations (1) and (2) into a single regression model where all the explanatory variables are interacted with the ideology of the government in year $t - 1$. The coefficient estimates for this combined model are included in table 9 in the appendix. Since dividing the sample makes the regression results much easier to read, I concentrate on these results in the main text of the article.

When it comes to control variables, it is especially important to control for factors that might potentially affect both the state of the economy in the year of the election ($t - 1$) and the ideology of the government in the following year (t), since they potentially render the relationship between downturns and ideological change spurious. The most obvious candidate is international conflict, which is likely to be associated with an increased risk of an economic downturn—especially when it comes to aggregate consumption (Barro and Ursúa 2008, 1)—

and with a change in the likelihood of a change in government. I deal with this complication in two ways. First of all, I exclude the two world wars (and their immediate aftermath) from the analysis entirely. Second, I control for participation in international military conflicts using data from the Correlates of War project (Sarkees and Wayman 2010).

I also control for several background variables that might presumably affect both the underlying likelihood of economic downturns and the underlying likelihood of ideological shifts. The first of these variables is presidentialism, which affects both how countries respond to economic events (Persson and Tabellini 2003) and, obviously, how they appoint their heads of government. The data on presidentialism come from Przeworski et al. (2013), with the exception that I count semi-presidential systems as parliamentary since my data on the ideology of heads of government relates to the prime minister, not the president, in systems with a dual executive.⁹ The second background variable is the electoral system, which is another institution that potentially affects both economic outcomes and the way in which heads of government are appointed. I include a dummy for proportional electoral systems that is based on the historical overview provided in Colomer (2004). The third background variable is economic development (a measure of real GDP per capita from Maddison [2011]). As additional background variables, I include two period dummies (for the pre-First World War and interwar periods, the reference category being post-Second World War) and a regional dummy for Latin America (the reference category being Western Europe, North America, and the Asia-Pacific region). The main reason for the inclusion of a dummy for Latin America is Latin America’s history of regime transitions—democratic breakdowns are almost always associated with downturns (Przeworski et al. 2000, 109)—and Latin America’s sharp turn to the left in the 1990s and 2000s, which influences the baseline likelihood of ideological shifts to the left in that particular region.

Main results

The results of the main statistical analyses can be found in table 3. The first two columns present results for shifts to the right (eq. [1] above). The last two columns present results for shifts to the left (eq. [2]).

9. In semi-presidential systems, such as France, the prime minister is typically responsible for domestic policy when the prime minister and the president represent different parties.

Table 3. Main Results

	Government at $t - 1$			
	Left or Centrist		Right	
	(1)	(2)	(3)	(4)
Short downturn	1.15*** (.26)	1.09*** (.35)	.08 (.33)	-.09 (.34)
Long downturn	.96* (.50)	.99** (.44)	1.27** (.52)	.95 (.65)
War		.58 (.55)		.18 (.50)
Presidentialism		.72* (.37)		.27 (.39)
Proportional electoral system		-1.18** (.50)		.67* (.39)
GDP per capita		.05 (.04)		.05 (.04)
Before First World War		-.34 (.84)		.91 (.72)
Interwar period		.69 (.62)		.66 (.61)
Latin America		.51 (.76)		1.10* (.63)
Observations	307	305	249	245
Countries	25	25	27	27

Note. Dependent variable: change in government ideology (0,1). Logit coefficients. Robust standard errors clustered by country in parentheses.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

Models 1 and 3, which do not include any control variables, simply reproduce the results in table 2 in regression form. Short economic downturns are associated with a large increase in the likelihood of a shift to the right if a left-wing or centrist government is in power, but they are only associated with a small and imprecisely estimated increase in the likelihood of a shift to the left if a right-wing government is in power. Long economic downturns, by contrast, are associated with an increased likelihood of ideological change regardless of the ideological orientation of the head of government at the time of the election (note that these coefficients are not very precisely estimated since long economic downturns are rare events).

Models 2 and 4 show that these findings are robust to the inclusion of control variables. In fact, the differences

between left-wing and centrist governments on the one hand (model 2) and right-wing governments on the other (model 4) are starker when the controls are included, for in model 4, the estimated effect of a short economic downturn on the likelihood of a shift to the left is negative; the estimated effect of a short economic downturn on the likelihood of a shift to the right, meanwhile, remains large and positive.¹⁰

Table 4 provides a substantive interpretation of the results reported in columns 2 and 4 in table 3. The numbers in table 4 represent the increase in the predicted probability of a change in government ideology in the wake of short and long downturns, conditional on the ideology of the government in the beginning of the election year (year $t - 1$). As I mentioned above, to compare the magnitudes of the marginal effects of the explanatory variables properly, it is necessary to estimate an interactive model that combines equation (1) and equation (2) into a single model. The predicted probabilities in table 4 are based on such a model.

The interpretation is straightforward: long downturns, lasting more than one year, appear to have very similar effects on different kinds of governments (although the effects are estimated imprecisely), but short downturns only increase the likelihood of shifts to the right (where left-wing or centrist governments are in power), not the likelihood of shifts to the left (where right-wing governments are in power). The difference between the estimated effects of short downturns on shifts to the left and shifts to the right is statistically significant ($p \approx .02$); the difference between the estimated effects of long downturns is not.

Duration and depth

To understand the political consequences of different kinds of economic downturns better, we need to consider how the effects of the duration of downturns combine with the ef-

10. One possible confounder is that various types of economic downturns might be more frequent under some governments than others. If right-wing governments pursue more contractionary fiscal policies, for example—as Herbert Hoover’s administration did in the beginning of the Great Depression—then right-wing governments might prolong economic downturns, turning “short” downturns into “long” downturns, which would explain the pattern that I find in this article. The differences between left and centrist governments on the one hand and right-wing governments on the other are in fact small, however. The (in-sample) proportion of short downturns where left and centrist governments were in power was 12.1% and the proportion of long downturns was 6.0%. Under right-wing governments, the proportion of short downturns was 13.4% and the proportion of long downturns was 6.7%. The fact that these proportions are so similar suggests that the likelihood of being in a (particular type of) downturn is not a function of the government’s ideology.

Table 4. Estimated Effects (I)

	Type of Downturn	
	Short	Long
Estimated increase in the likelihood of a shift to the right if government at $t - 1$ is left-wing or centrist	21.3***	19.1**
Estimated increase in the likelihood of a shift to the left if government at $t - 1$ is right-wing	-1.9	21.9

Note. The predicted probabilities, in percent, are calculated on the basis of an interactive specification that combines models 2 and 4 in table 3 into a single regression model. The estimates from the combined model are included in table 9 in the appendix.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

fects of their depth. I expect downturns that are both deep and long-lasting to have economic effects that are sufficiently widespread to lead to an increased likelihood of a shift to the left; in all other cases, I expect that recessions will be more harmful for center-left parties than for right-wing parties.

As table 5 shows, this has been the case historically. To examine the combined effects of duration and depth, I have estimated models that include four dummies for different kinds of economic downturns: one dummy for short and shallow downturns, one for short and deep downturns, one for long and shallow downturns, and one for long and deep downturns. "Shallow" means, here, that the drop in consumption per capita was smaller than the median (for short and long downturns, respectively) and "deep" means that the drop in consumption per capita was greater than the median. The estimates in table 5 suggest that the likelihood of a shift to the right (when a center-left government is in power) increases in all kinds of downturns, whereas the likelihood of a shift to the left (when a right-wing government is in power) only increases significantly when downturns are both long-lasting and deep.

Like table 4, table 6 provides a substantive interpretation of the regression results. As the table reveals, deep, long-lasting downturns are associated with a high likelihood that right-wing governments are replaced by center-left governments, but where center-left governments are in power, the estimated effect of deep, long downturns is much weaker. Conversely,

short downturns (and long, mild downturns) are associated with a significant increase in the likelihood that center-left governments are replaced by right-wing governments but appear to do little harm to right-wing parties.

Robustness checks

The robustness checks that are reported in the appendix show, among other things, that the results vary only slightly

Table 5. The Depth of Downturns

	Government at $t - 1$			
	Left or Centrist		Right	
	(1)	(2)	(3)	(4)
Short, shallow downturn	1.08*** (.39)	.89** (.44)	.05 (.44)	-.02 (.48)
Short, deep downturn	1.22*** (.38)	1.27** (.51)	.12 (.43)	-.17 (.41)
Long, shallow downturn	1.18* (.66)	1.35** (.55)	.68 (.86)	.04 (.99)
Long, deep downturn	.67 (.65)	.52 (.68)	1.78** (.86)	1.63 (1.01)
War		.59 (.56)		.17 (.51)
Presidentialism		.72* (.38)		.29 (.39)
Proportional electoral system		-1.19** (.50)		.74* (.40)
GDP per capita		.05 (.03)		.05 (.04)
Before First World War		-.32 (.83)		.93 (.71)
Interwar period		.69 (.63)		.62 (.61)
Latin America		.54 (.76)		1.02 (.63)
Observations	307	305	249	245
Countries	25	25	27	27

Note. Dependent variable: change in government ideology (0,1). Logit coefficients. Robust standard errors clustered by country in parentheses.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

Table 6. Estimated Effects (II)

	Type of Downturn			
	Short, Shallow	Short, Deep	Long, Shallow	Long, Deep
Estimated increase in the likelihood of a shift to the right if government at $t - 1$ is left-wing or centrist	16.9*	25.2**	27.0**	9.2
Estimated increase in the likelihood of a shift to the left if government at $t - 1$ is right-wing	-.3	-3.5	.8	36.4*

Note. The predicted probabilities are calculated on the basis of an interactive specification that combines models 2 and 4 in table 3 into a single regression model. The estimates from the combined model are included in table 9 in the appendix.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

when long and short economic downturns are defined differently (table 7; of particular interest are the analyses that include continuous measures of long-term and short-term growth), when controlling for domestic disorder and when using a different definition of democracy (table 8), when controlling for party fractionalization (table 10), and when the first years of the Great Depression and the Great Recession—that is, the observations that were included in Lindvall [2014])—are excluded from the analysis (table 12).

I have checked for parameter stability between regions and over time by estimating separate downturn effects for different regions and periods (table 13 in the appendix). Dividing the sample into subsamples is costly, since there are few downturn events in the data to begin with (especially when it comes to long downturns), but I would like to comment on three findings: first of all, the finding that “long” recessions are harmful for the political right does not apply in Latin America; second, economic downturns do not appear to have had very strong effects before the First World War (possibly because downturns were more frequent then); third, the evidence for the claim that long recessions are associated with a higher likelihood of a change in government ideology is strongest in data from the interwar period. To check for parameter stability across institutional contexts, I have also estimated separate downturn effects for presidential and parliamentary

systems, and for majoritarian and proportional electoral systems (table 14). Just like the region- and period-specific findings, these findings need to be interpreted with caution, but they do suggest that the effects of recessions are generally weaker in proportional systems than in majoritarian systems, and they suggest that long recessions are particularly harmful for the right in parliamentary systems.

CONCLUSION

This article has used new data on the ideological orientation of heads of government in 27 countries over a period of almost 140 years—between the early 1870s and 2010—to study the relationship between economic downturns and the struggle for political power between parties on the left and parties on the right. The main finding is that in a “short” economic downturn, lasting no more than one year, a shift to the right is significantly more likely than a shift to the left. This finding is broadly consistent with the “swing-to-the-right hypothesis” that scholars such as Alt (1979), Durr (1993), and Stevenson (2001, 2002) have explored in earlier work (using very different sorts of data). But the article also provides evidence that long economic downturns are different from short downturns: whereas short downturns favor the political right, long downturns appear to be associated with an increased likelihood of ideological shifts in both directions (in other words, they render right-wing governments just as vulnerable as centrist or left-wing governments).

There are numerous ways in which the arguments in this article could be extended. First of all, one of the main ideas of this article—that short economic downturns and long economic downturns have different political effects—has not been tested systematically using micro-level data. The macro-level results that I report here are suggestive, and reasonably robust, but it should also be possible to test the same set of hypotheses with survey evidence. The economic crisis that began in 2007–8, which resulted in a brief (but deep) economic downturn in some countries but led to more long-lasting economic downturns in other countries, is a promising test case. Second, I have concentrated on shifts to the left and shifts to the right that are associated with elections (events during years when no elections were held have not been included in the sample) since this paper’s arguments are concerned with what happens in elections. But some studies of executive politics and cabinet formation suggest that economic factors may also play a role in changes in political leadership that occur between elections (Saalfeld 2008). One possible extension of the analyses in this article is to develop a model of both election-induced and non-election-induced changes in the identity and ideology of heads of government.

ACKNOWLEDGMENTS

I am grateful to Larry Bartels, Pablo Beramendi, Thomas Brambor, and Yvette Peters for their comments on earlier versions of this article, and to Randolph Stevenson for helpful advice. I also thank the editor and three anonymous reviewers for their feedback and suggestions. Previous versions of this manuscript were presented at Duke University, Harvard University, the workshop on Economic Inequality and the Quality of Democracy at the University of British Columbia (February 7–8, 2014), at the 2014 American Political Science Association meeting, and at the 2015 meeting of the European Political Science Association.

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